

30-Minute Student Task

Build Your Own Rule-Based Python Chatbot

Name	KANIKARAM MEDHARA POOJITHA
Registration number	99230040116
Programming Language	Python

ASSESSMENT

A simple rule-based conversational chatbot was developed in Python. The chatbot accepts user input continuously, recognizes greetings and different questions, responds accordingly, and exits when the user types bye or exit.

INTRODUCTION

The Student Management System is a Python-based application developed to manage student records in a simple and efficient way. The system allows the user to add, view, search, update, and delete student information.

This project demonstrates important Python programming concepts such as dictionaries, loops, conditional statements, input handling, and CRUD operations. It provides a simple menu-driven interface that makes the system easy to understand and use.

OBJECTIVE

The main objectives of the Student Management System are:

- To maintain student records efficiently.
- To add new student information.
- To display stored student records.
- To search for a particular student.
- To update existing student information.
- To delete student records when required.
- To demonstrate practical implementation of Python programming concepts.

PYTHON SOURCE CODE

```
students = {}

while True:
    print("\n===== STUDENT MANAGEMENT SYSTEM =====")
    print("1. Add Student")
    print("2. View Students")
    print("3. Search Student")
    print("4. Update Student")
    print("5. Delete Student")
    print("6. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        roll = input("Enter Roll Number: ")
        name = input("Enter Name: ")
        marks = input("Enter Marks: ")

        students[roll] = {
            "Name": name,
            "Marks": marks
        }
        print("Student Added Successfully!")

    elif choice == "2":
        if not students:
            print("No Student Records Found")
        else:
            print("Student Records")
            for roll, data in students.items():
                print("Roll No:", roll)
                print("Name:", data["Name"])
                print("Marks:", data["Marks"])
                print("-----")

    elif choice == "3":
        roll = input("Enter Roll Number to Search: ")

        if roll in students:
            print("Student Found")
            print("Name:", students[roll]["Name"])
            print("Marks:", students[roll]["Marks"])
        else:
            print("Student Not Found")

    elif choice == "4":
        roll = input("Enter Roll Number to Update: ")

        if roll in students:
            name = input("Enter New Name: ")
            marks = input("Enter New Marks: ")
            students[roll]["Name"] = name
            students[roll]["Marks"] = marks
            print("Record Updated Successfully!")
        else:
            print("Student Not Found")

    elif choice == "5":
        roll = input("Enter Roll Number to Delete: ")

        if roll in students:
            del students[roll]
            print("Student Deleted Successfully!")
        else:
            print("Student Not Found")

    elif choice == "6":
        print("Thank You!")
        break

    else:
        print("Invalid Choice")
```

Chatbot Execution and Test Output

1) Add Student

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 1
Enter Roll Number: 99230040116
Enter Name: KANIKARAM MEDHARA POOJITHA
Enter Marks: 85
Student Added Successfully!
```

2) View Student

```
===== STUDENT MANAGEMENT SYSTEM =====
Enter your choice: 2

Student Records
Roll No: 99230040116
Name: KANIKARAM MEDHARA POOJITHA
Marks: 85
-----
```

3) Search Student

```
===== STUDENT MANAGEMENT SYSTEM =====
Enter your choice: 3
Enter Roll Number to Search: 99230040116
Student Found
Name: KANIKARAM MEDHARA POOJITHA
Marks: 85
```

4) Update Student

```
===== STUDENT MANAGEMENT SYSTEM =====  
Enter your choice: 4  
Enter Roll Number to Update: 99230040116  
Enter New Name: KANIKARAM MEDHARA POOJITHA  
Enter New Marks: 90  
Record Updated Successfully!
```

5) Delete Student

```
===== STUDENT MANAGEMENT SYSTEM =====  
Enter your choice: 5  
Enter Roll Number to Delete: 99230040116  
Student Deleted Successfully!
```

RESULT

The Student Management System was successfully developed using Python. The application successfully performs student record management operations such as adding, viewing, searching, updating, and deleting student information. The project demonstrates the practical implementation of basic Python programming concepts in a real-world management application.